

Exact Periodic Orbits and Algebraic Zeta Geometry of One-Vertex Edge-Type Dyck Shifts

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Abstract

We close the periodic-point theory of the edge-type Dyck shift associated with one vertex and N loop edges. The algebraic zeta function yields explicit odd and even fixed-count formulas for every N and every period, followed by primitive points and orbits through Möbius inversion. We distinguish the $N = 1$ full-two-shift boundary from the $N > 1$ double dominant pole, quadratic branchpoints, nonrationality, and orbit asymptotics. Formal series, symbolic algebra, and direct periodic-word enumeration independently audit the result; the source-locked entropy is $\log(N + 1)$.

1 Model and algebraic zeta

Let D_N^E be the edge-type Dyck shift for the graph with one vertex and N loop edges. Its alphabet is $\{a_i, b_i : 1 \leq i \leq N\}$, with inverse-monoid relation $a_i b_j = 1$ for $i = j$ and 0 otherwise. A finite block is admissible when its reduction is nonzero. Put

$$s_N(z) = \sqrt{1 - 4Nz^2}.$$

The context-free circular-code series satisfies

$$g_N(z) = \frac{Nz^2}{1 - g_N(z)}, \quad g_N(z) = \frac{1 - s_N(z)}{2},$$

where the second expression is the small solution. The source theorem, specialized to the one-vertex N -loop graph, gives

$$\zeta_N(z) = \frac{1 - g_N(z)}{(1 - Nz - g_N(z))^2}.$$

Substitution, rather than a claim of priority, yields

$$\zeta_N(z) = \frac{2(1 + s_N(z))}{(1 + s_N(z) - 2Nz)^2}. \quad (1)$$

Theorem 1 (all fixed and primitive counts). *Let $F_N(n) = |\text{Fix}(\sigma^n)|$. If n is odd, then*

$$F_N(n) = 2 \left((N + 1)^n - \sum_{j=0}^{(n-1)/2} \binom{n}{j} N^j \right).$$

If n is even, then

$$F_N(n) = 2 \left((N + 1)^n - \sum_{j=0}^{n/2} \binom{n}{j} N^j \right) + \binom{n}{n/2} N^{n/2}.$$

The numbers of least-period points and primitive orbits are

$$P_N(n) = \sum_{d|n} \mu(n/d) F_N(d), \quad C_N(n) = P_N(n)/n.$$

Proof. By definition, $z\zeta'_N(z)/\zeta_N(z) = \sum_{n \geq 1} F_N(n)z^n$. Insert (1), expand the quadratic series $s_N(z)$, and separate the central binomial term when n is even. The two displayed parity formulas result. Möbius inversion isolates least periods. A least-period- n orbit has exactly n possible origins, proving the last division. $\square$

Source note

The model, zeta formula, and entropy lock follow Krieger–Matsumoto [1]; Keller’s loop-counting framework gives related context [2]. The official Münster volume PDF controls pagination; an arXiv journal-reference entry reports 171–185. No priority claim is made.

References

- [1] W. Krieger and K. Matsumoto, Zeta functions and topological entropy of the Markov-Dyck shifts, *Münster J. Math.* 4 (2011), 171–184, arXiv:0706.3262.
- [2] G. Keller, Circular codes, loop counting, and zeta-functions, *J. Combin. Theory Ser. A* 56 (1991), 75–83, doi:10.1016/0097-3165(91)90023-A.

Declarations

Scope. The exact registered literal is NO_BAD_EULER_OR_ROOT_NUMBER. **Data and code.** Exact ledgers, independent checks, and build instructions are supplied in the HCS-C205 release package. **Competing interests.** None declared. **AI-use disclosure.** Generative AI tools assisted drafting and code generation. Every theorem statement, source record, exact output, and scope claim was checked through the released internal artifact audit chain; this is not external peer review or an independent error process.